Shreyas Sunil Gaikwad

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Applied mathematician, deep learning researcher, and software engineer with experience in high-dimensional Bayesian inversion and uncertainty quantification, automatic differentiation, differentiable simulations, gradient-based optimization, physics-informed deep learning, generative AI (GenAI), and production C++.

EDUCATION

The University of Texas at Austin
Ph.D. in Computational Science, Engineering, and Mathematics (CSEM), GPA: 4.0/4.0 — *2020–2025*
M.S. in Computational Science, Engineering, and Mathematics (CSEM), GPA: 4.0/4.0 — *2019–2021*

Indian Institute of Technology (IIT) Bombay, India
B.Tech with Honors in Mechanical Engineering (Minor in Computer Science), Rank: 4/150, GPA: 9.32/10.0 — *2015–2019*

WORK EXPERIENCE

- **Lead Software R&D Engineer, Cadence Design Systems** — *2026–Present*
  - Contributing to **C++** platform software for a **multi-decabillion-gate** hardware emulation system (Palladium/Protium) used to validate computer chip designs before physical manufacturing
  - Diagnosing and resolving urgent customer-reported issues (**P0/P1**), including subtle edge cases in concurrent, non-deterministic logic, for high-performance, low-latency graph-traversal code underlying real-time hardware emulation
  - Pruned bad placements, **cutting emulator scheduling trials 50%** within 2% of best step count via 3 ML models (**pairwise learning-to-rank**) and a domain heuristic on engineered scale-invariant features, in a low-data regime
  - Exploring use of Graph Neural Networks (**GNNs**) paired with ML to improve Palladium emulation performance
  - Developing Google Test-based unit test suite of **200+** cases covering optimization logic and multi-threaded task-bucketing
- **Graduate Research Assistant, UT Austin** — *2020–2025*
  Ph.D. Thesis: Paleo-reconstruction of Greenland Ice Sheet Dynamics over the Holocene via Bayesian Inversion
  - Engineered the Fortran-based ice sheet model SICOPOLIS into a **differentiable simulation** using open-source **Algorithmic Differentiation (AD)**, enabling scalable **Bayesian inversion** with efficient adjoint-based gradients [3]
  - Solved high-dimensional inverse problems constrained by partial differential equations (PDEs), estimating $\mathcal{O}(\mathbf{10^6})$ uncertain parameters in a continental-scale ice sheet model, accelerated using preconditioned quasi-Newton methods
  - Reduced sensitivity analysis cost from $\mathcal{O}(\mathbf{N^3})$ to $\mathcal{O}(\mathbf{1})$ model evaluations using adjoint-based gradients
  - Developed a novel interpretation technique for neural networks using physics-informed adjoints and **eXplainable-AI (XAI)** methods in **Keras**, validating learned insights against physically consistent adjoint sensitivities [1]
- **Machine Learning Intern, Ansys Inc.** — *Summer 2024*
  - Built a unified **Neural Radiance Field (NeRF)** architecture to implicitly represent multiple, diverse object geometries
  - Developed a scalable **GenAI** pipeline for image-to-3D synthesis in **PyTorch** (**106 million parameters**), combining the unified NeRF architecture with a pre-trained, 3D-aware, **multi-view 2D diffusion model** for 3D asset generation
  - Trained the NeRF without 3D data using 3D-aware diffusion priors and **Score Distillation Sampling (SDS)** loss
- **Visiting Scholar, Argonne National Laboratory** — *Summer 2022*
  - Developed the **first open-source**, **MPI**-parallel, AD-based, adjoint-enabled Bayesian optimization framework for atmospheric and ocean general circulation model MITgcm; alternative to a proprietary tool (**$15,000/year/user**) [2]

RESEARCH PROJECTS

- **Physics-Informed Machine Learning** — *Spring 2021*
  - Trained a deep neural network in **PyTorch** as a differentiable emulator of a nonlinear PDE-based mountain glacier model, assimilating data via higher-order derivatives from PyTorch's computational graph
- **Laplacian 2D Finite Difference (FD) Solver Application** — *Fall 2020*
  - Features: OOP (**C++**); solvers (Gauss, Jacobi, PETSc); build (Autotools); testing (Bats, Travis CI, Docker); **98% coverage** (lcov); **0% memory errors** (Valgrind); HPC environment (SLURM); parsing & logging (GRVY)

TECHNICAL SKILLS

- **Languages**: Python (PyTorch, Keras, TensorFlow, JAX), C/C++, Julia, Fortran-77/90, MATLAB
- **HPC toolkit**: OpenMP, MPI, CUDA, SLURM, git, docker, shell, CI, autotools, valgrind, lcov, GRVY, Tapenade (AD)

PUBLICATIONS & OPEN-SOURCE CONTRIBUTIONS

1. **S.S. Gaikwad***, H. Pillar*, D. Apte* et al., "Pairing eXplainable AI with Adjoint Modeling for Flexible Investigation and Robust Attribution of Ocean Variability", *In preparation for Geophysical Research Letters (2026).*
2. **S.S. Gaikwad** et al., "MITgcm-AD v2: Open source tangent linear and adjoint modeling framework for the oceans and atmosphere enabled by the AD tool Tapenade", *Future Generation Computer Systems (2024).* [code]
3. **S.S. Gaikwad** et al., "SICOPOLIS-AD v2: tangent linear and adjoint modeling framework for ice sheet modeling enabled by automatic differentiation tool Tapenade", *Journal of Open Source Software (2023).* [code]
4. diffblas, open-source differentiated BLAS routines from their implementation in LAPACK using AD tool Tapenade

* *Equal contributions to this manuscript.*